Maulana Azad National Institute of Technology, Bhopal
Department of Electrical Engineering

Class: B.Tech. IIV Sem. Sub: Generation of Electrical Power (E-224)

Session- Jan-April. 2025

Assignments-1

1. Provide the data of Installed Generation capacity in India in following format. Also compile the same data State wise.

Installed Capacity in India(as on 30-11-2024)		
Type of plant and fuel used	**Total installed generation capacity (MW) for each type of plant and fuel wise**	**% of Total installed Generation capacity (For each individual type of power plant and fuel used)**
Total Thermal Generation		
Coal		
Lignite		
Gas		
Diesel		
Total HYDRO Based		
Total NUCLEAR Based		
Total RES Based		
Wind Based		
Solar Based		
Waste to Energy Based		
Bagasse Cogeneration/Biomass Power & Gasification		
Total Installed Capacity in India including all type of plants (MW)		

2. The load on a power plant on a typical day as under. Plot the chronological load curve, load duration curve and calculate the load factor of the plant and the energy supplied by the palnt in 24 hours. Also draw the integrated load duration curve and mass curve.

Time	12-5 AM	5-9	9-6 PM	6PM-10PM	10PM-12Mid Night
Load (MW)	20	40	80	100	20

3. The yearly load duration curve of a certain power station can be considered as a straight line from 3000 MW to 80 MW. Power is supplied with one generator unit of 200 MW capcity and two units of 100 MW capacity each. Determine (i) Installed capacity (ii) Load factor (iii) Plant factor (iv) Max. Demand (v) utilization factor.

4. A central station supplies energy to two sub stations. 4 feeders take off from each of the sub-stations. The maximum demands are as under:

Central Station	10 MW
Sub-Station A	6 MW
Sub-Station B	8 MW
Feeders on sub-station A	1.5, 2,5,3 MW
Feeders on sub-station B	2,4,5,1 MW

Calculate the diversity factor between

(a) Sub-stations

(b) Feeder of sub-station A

(c) Feeders of sub-station B

5. A feeder supplies 3 distribution transformers which feed the following connected loads:

Transformer 1: Motor loads 30 kW, demand factor 0.6, Commercial loads 10 kW, demand factor 0.5

Transformer 2: Residential loads 50 kW, Demand factor 0.4.

Transformer 3: Residential loads 50 kW, Demand factor 0.5.

The diversity factor for the loads on the three transformers may be taken as 1.8, 2.5 and 3. The diversity factor between transformers may be taken as 1.1. Find (a) Peak load on each transformer (b) Peak load on the feeder

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